Tides

OBJECTIVE

In this lesson, students will learn what causes high tides and low tides on Earth. They will discover how the gravitational fields of the Moon and Sun affect Earth's tides.

ACARA CONTENT DESCRIPTIONS

Predictable phenomena on Earth, including seasons and eclipses, are caused by the relative positions of the sun, Earth and the moon (ACSSU115)

- investigating how natural phenomena such as phases of the moon can cause tides
- modelling the relative movements of the Earth, sun and moon

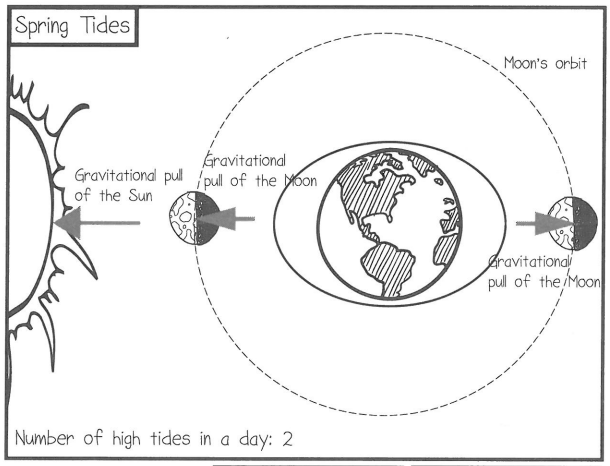
LESSON PLAN

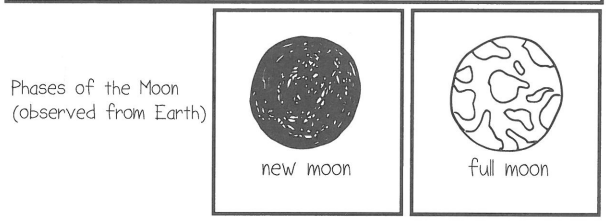
Activities	Resources
Activity 1: Spot the Difference Open the presentation to the first slide which is an image of Le Mont-Saint-Michel. There is an obvious difference between the sea levels in the two pictures. Ask students what they already know about tides and why they occur. 10	• Presentation: Tides
Activity 2: Tide Away! Play the video to students. Once the video has finished, give out the Tide Away! 1 and 2 worksheets, scissors and glue. While students are cutting out the shapes on Tide Away! 1, play the video again. This activity requires students to create diagrams of spring and neap tides. This might be tricky for some students, so offer guidance if required. Once students have completed their diagrams, use slides 3-6 of the presentation to review answers so students can self-mark their own work. 40	• Presentation: Tides • ClickView video *Cycles and Season* Chapter 6: **https://clickv.ie./w/lp/13** • Photocopies of the Tide Away! 1 and 2 worksheets • Scissors and glue
Activity 3: Why Do We Have to Know about Tides? Using the last slide of the presentation, discuss as a class reasons why it is important for people to understand tides and their movements. Possible answers could include: • Helps navigation at sea • Important in the construction of buildings, jetties, docks, boardwalks etc. • Helps fishermen improve their catches 10	• Presentation: Tides

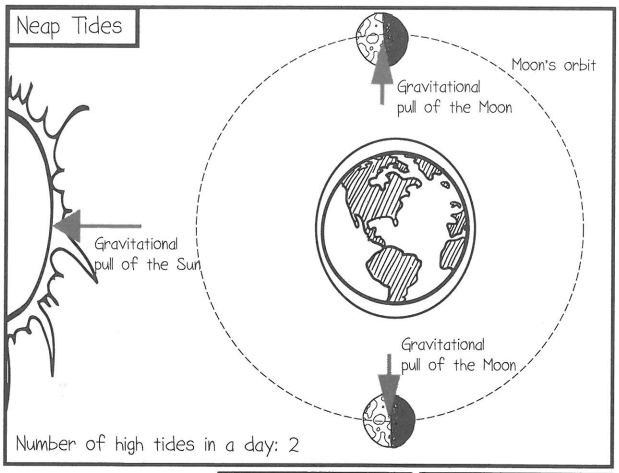


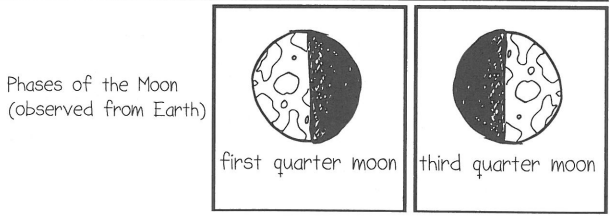
ANSWERS

Tide Away!









EARTH AND SPACE SCIENCES

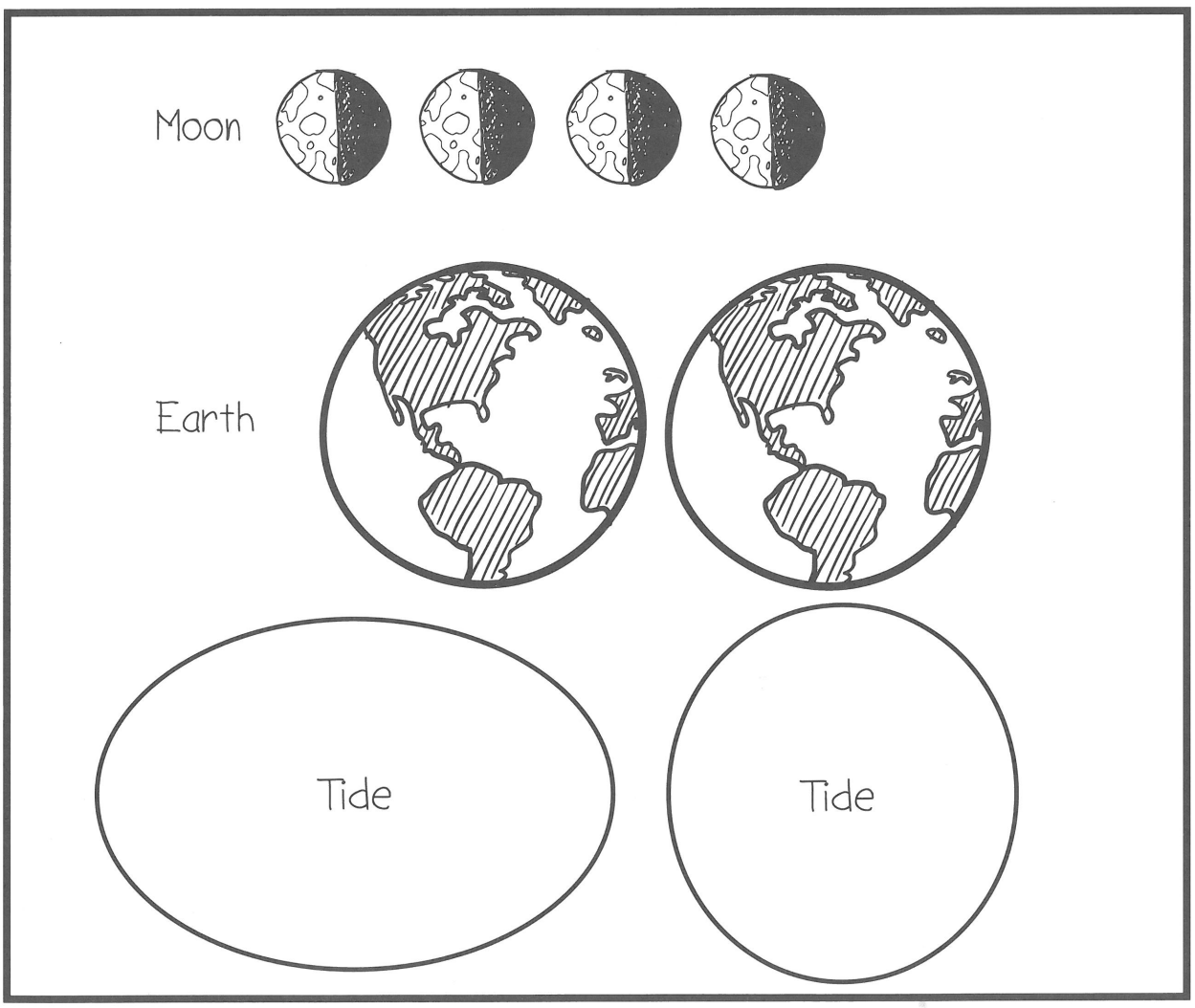
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Tide Away! 1

Instructions:

1. Watch Chapter 6 of the video: *Cycles and Seasons.*
2. Cut out the shapes in Box A.
3. Glue the tide that represents a spring tide over the * symbol on the Tide Away! 2 worksheet.
4. Glue the tide that represents a neap tide over the * symbol on the Tide Away! 3 worksheet.
5. Glue Earth over each tide.
6. Position the moons on the diagram to show their gravitational pull on each of the tides.
7. Draw arrows to show the gravitational pull of the Moon and the Sun.
8. Draw the phases of the Moon that would be seen from Earth during each of the tides.
9. Determine how many high tides you will experience in one day. (Hint: How long does it take Earth to rotate on its axis?)

Box A

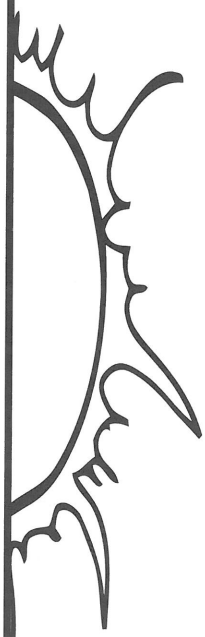


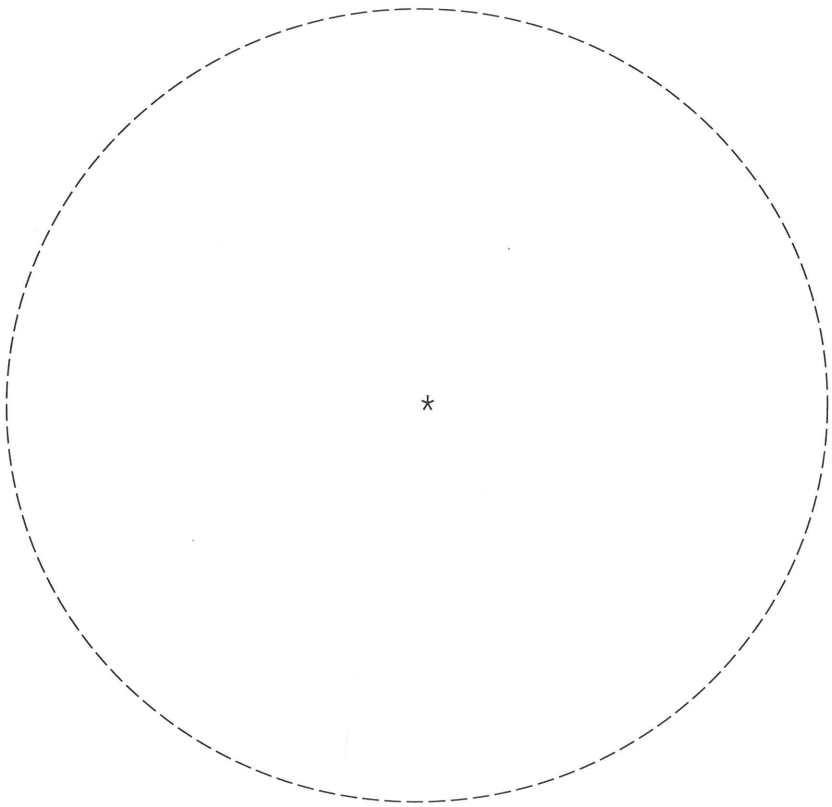
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Tide Away! 2 - Spring Tides

Glue the shapes from Tide Away! 1 on the diagram below to represent spring tides. Include arrows to show the direction of the gravitational pull of the Sun and the Moon.

Spring Tide





Number of high tides in a day: _____

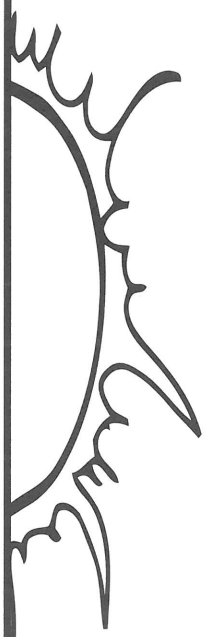
Draw the phases of the Moon you would see from Earth during a spring tide.

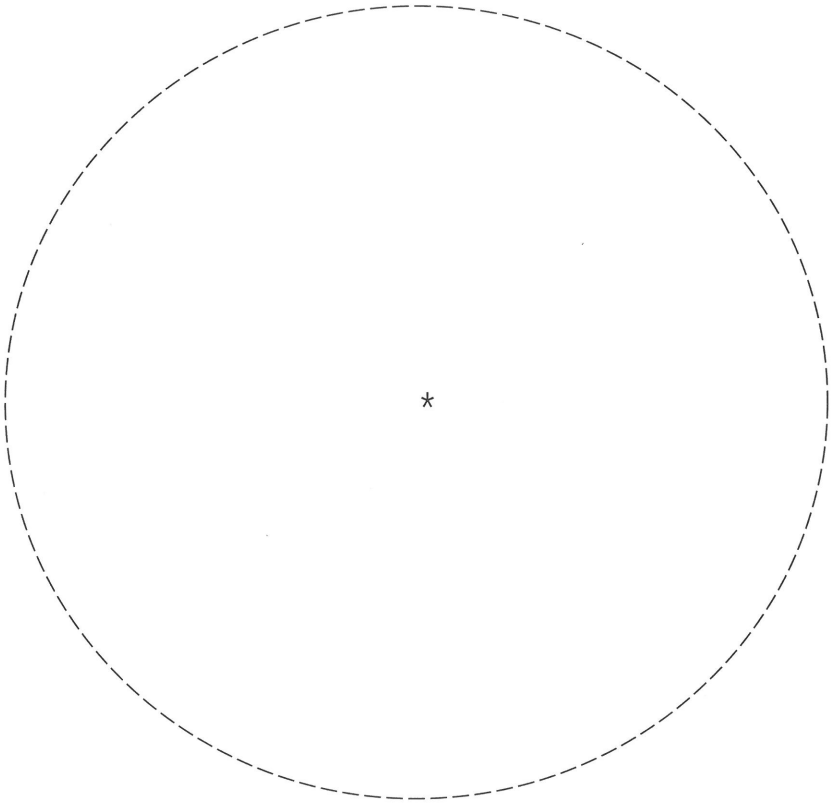
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Tide Away! 3 - Neap Tides

Glue the shapes from Tide Away! 1 on the diagram below to represent neap tides. Include arrows to show the direction of the gravitational pull of the Sun and the Moon.

Neap Tide





Number of high tides in a day: _____

Draw the phases of the Moon you would see from Earth during a neap tide.

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